6. In a G.P. the 9th term is 32805 and 6th term is 1215. Find the 12th term.
7. Find the 10th term of a G.P. whose 8th term is 768 and the common ratio is 2.
8. If $a,\ b,\ c$ are in A.P. then show that $3^a, 3^b, 3^c$ are in G.P.
9. In a G.P. the product of three consecutive terms is 27 and the sum of the product of two terms taken at a time is $\frac{57}{2}$. Find the three terms.
10. A man joined a company as Assistant Manager. The company gave him a starting salary of ₹$60{,}000$ and agreed to increase his salary 5% annually. What will be his salary after 5 years?
11. Sivamani is attending an interview for a job and the company gave two offers to him.

    Offer A: ₹$20{,}000$ to start with followed by a guaranteed annual increase of 6% for the first 5 years.

    Offer B: ₹$22{,}000$ to start with followed by a guaranteed annual increase of 3% for the first 5 years.

    What is his salary in the 4th year with respect to the offers A and B?
12. If a, b, c are three consecutive terms of an A.P. and x, y, z are three consecutive terms of a G.P. then prove that $x^{b-c} \times y^{c-a} \times z^{a-b} = 1$.

2.10 Sum to *n* terms of a Geometric progression

A series whose terms are in Geometric progression is called Geometric series.

Let $a,\ ar,\ ar^2, \ldots ar^{n-1}, \ldots$ be the Geometric Progression.

The sum of first n terms of the Geometric progression is

$$S_n = a + ar + ar^2 + \cdots + ar^{n-2} + ar^{n-1} \quad \ldots (1)$$

Multiplying both sides by r, we get $rS_n = ar + ar^2 + ar^3 + \cdots + ar^{n-1} + ar^n \quad \ldots (2)$

$$(2)-(1) \Rightarrow rS_n - S_n = ar^n - a$$

$$S_n(r-1) = a(r^n - 1)$$

Thus, the sum to n terms is $S_n = \frac{a(r^n - 1)}{r-1},\ r \neq 1$.

Note

The above formula for sum of first n terms of a G.P. is not applicable when $r = 1$.

If $r = 1$, then

$S_n = a + a + a + \cdots + a = na$

Progress Check

1. A series whose terms are in Geometric progression is called _____.
2. When $r = 1$, the formula for finding sum to n terms of a G.P. is _____.
3. When $r \neq 1$, the formula for finding sum to n terms of a G.P. is _____.

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